

Uniform Random Mapping Functional-Graph Closure: Exact Cycle–Component Counts and Marked-Orbit Limits

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Abstract

For a uniform random function $f : [n] \rightarrow [n]$, we give a self-contained uniform random mapping functional-graph closure. Decomposing the graph into a permutation and a rooted forest yields the exact joint count of cyclic vertices and components, its marginal, and every expected cycle count.

1 Model and theorem

Choose each of the n^n functions $f : [n] \rightarrow [n]$ with equal probability. Its directed functional graph has edges $v \rightarrow f(v)$. Let C_n be the number of cyclic vertices and K_n its number of weak components. Write $c(k, r)$ for the unsigned Stirling number of the first kind and $(n)_k = n(n-1) \cdots (n-k+1)$.

Theorem 1 (Finite and asymptotic closure). *For $1 \leq r \leq k \leq n$,*

$$\#\{f : C_n = k, K_n = r\} = \binom{n}{k} c(k, r) k n^{n-k-1}, \quad (1)$$

where the last factor means the unique empty forest when $k = n$. Hence

$$\mathbb{P}(C_n = k) = \frac{(n)_k k}{n^{k+1}}, \quad \mathbb{E}Z_{n,\ell} = \frac{(n)_\ell}{\ell n^\ell}. \quad (2)$$

where $Z_{n,\ell}$ counts directed ℓ -cycles.

These distributions are classical random-mapping statistics [1, 2]. Our contribution is a source-locked proof synthesis and executable boundary audit, not a literature-priority claim.

2 Permutation and forest proof

Fix the set S of k cyclic labels. The restriction to S is a permutation, and $c(k, r)$ permutations have r cycles. Every label outside S lies in an in-tree directed toward a root in S . The corresponding complete-graph Laplacian minor, of size $m = n - k$, is

$$L_S = nI_m - J_m.$$

For $m > 0$ its eigenvalues are k once and n with multiplicity $m - 1$, so $\det L_S = k n^{n-k-1}$. At $m = 0$ the empty determinant is one. Choosing S , its permutation, and this forest is bijective with the maps in (1).

Summing over r and using $\sum_r c(k, r) = k!$ gives

$$\#\{f : C_n = k\} = \binom{n}{k} k! k n^{n-k-1}.$$

After division by n^n , this is the first formula in (2). There are $\binom{n}{\ell}(\ell-1)! = (n)_\ell/\ell$ possible directed ℓ -cycles. Each prescribes ℓ independent function values and has probability $n^{-\ell}$, so indicator linearity proves the second formula. In particular, $n = 1$ gives the single fixed-point map, while $k = n$ reduces (1) to $c(n, r)$ as it must for permutations.

References

- [1] B. Harris, *Probability distributions related to random mappings*, Ann. Math. Statist. **31** (1960), 1045–1062, doi:10.1214/aoms/1177705677.
- [2] P. Flajolet and A. M. Odlyzko, *Random mapping statistics*, Advances in Cryptology—EUROCRYPT ’89, LNCS 434 (1990), 329–354, doi:10.1007/3-540-46885-4_34.